

Total No. of Questions: 08

Total No. of Printed Pages: 2
Roll No..

IT - 305
B. Tech. III Semester
Examination, December 2025
Grading System (GS)
Digital Circuits and Systems
Time: Three Hours

Maximum Marks: 70

Note: i) Attempt any five questions.
ii) All questions carry equal marks.
iii) Assume suitable data, if required.

1. a). Convert the following numbers as directed (any two):
 - i) $(9A)_{16}$ to Binary
 - ii) $(745)_8$ to Decimal
 - iii) $(1101011)_2$ to Hexadecimalb) Explain BCD, Excess-3 and Gray codes. Write their applications.
2. a) Explain Full Adder with truth tables and logic diagrams.
Design a full adder using two half adders..
b) Simplify the following Boolean function using K-map and implement it using NAND gates:
 $F(A, B, C) = \Sigma(0, 2, 3, 5, 7)$
3. a) Design a 4-to-16 decoder using 3-to-8 decoders and explain its operation.
b) Explain Shift Registers and their types with applications.
4. a) Explain Counters. Describe Johnson counter and Ring counter with timing diagrams.
b) Compare NMOS, PMOS and CMOS logic in detail.

5. a) Design a 2 input TTL NAND gate and explain its working.
b) Write a short note on 7-segment LED display.
6. a) Explain Multivibrators. Describe Monostable multivibrators with diagrams.
b) Explain Analog to Digital Converters. Describe any one ADC in detail.
7. a) What does edge triggered and level triggered mean?. Differentiate between Edge triggered and Level triggered flip-flops.
b) Draw AND, OR and NOT gates using NOR gate only.
8. Write a short note on (Any Four)
 - a) Draw 8×1 MUX using 2×1 Mux
 - b) Schmitt trigger circuit
 - c) Gray to binary code conversion